SOFTWARE ENGINEERING

LEVEL ONE

⇒ Introduction to software engineering:

What is Software Engineering?

Software engineering is the systematic process of, planning, designing, developing, testing, and maintaining software solutions. Much like construction, it requires meticulous planning and organization before implementation to ensure the final product is robust, scalable, and effective at solving real-world problems. While construction relies on physical materials like bricks, software engineering utilizes code to build digital infrastructure.

⇒ Who is a Software Engineer?

A software engineer is a professional who applies engineering principles and programming expertise to design and develop software. They transform abstract ideas into functional applications, such as websites, mobile apps, and enterprise systems. Their role encompasses the entire software development lifecycle—from conceptualization and coding to debugging, testing, and continuous maintenance—ensuring that the technology we rely on daily remains efficient, secure, and up-to-date.

Website sqi.edu.ng (Static information) browser

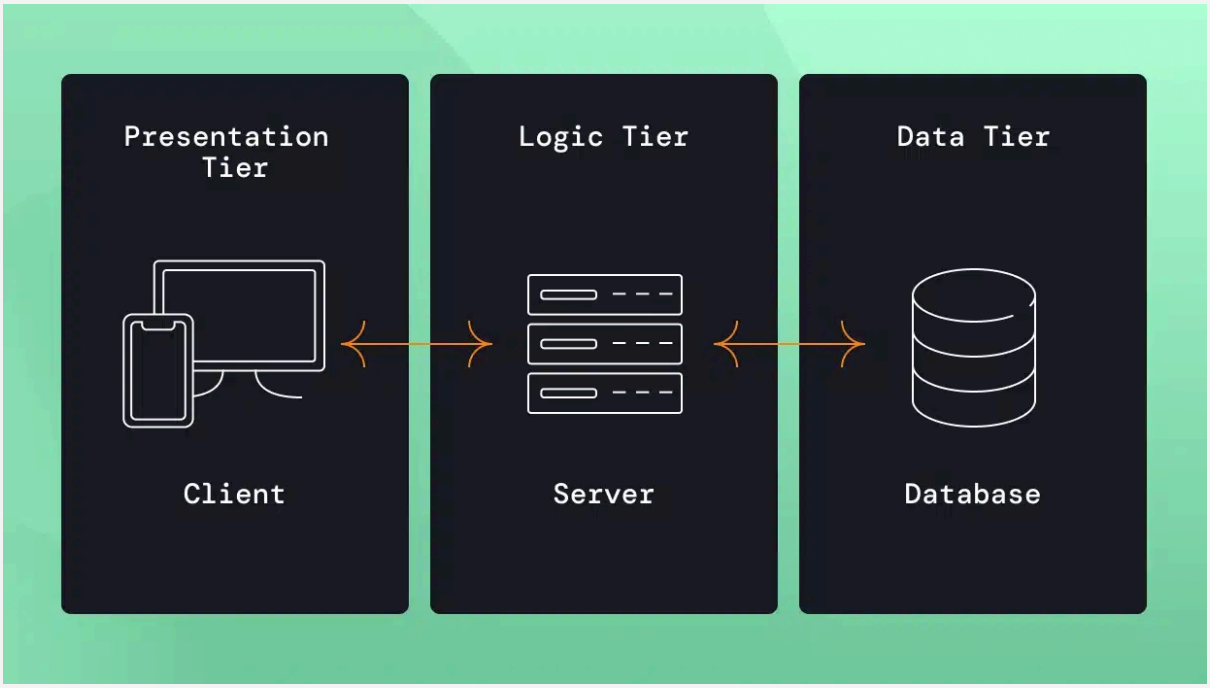
Web Applications jumia.com.ng scict.edossier.app/student-portal (Dynamic information person specific) browser

Mobile Applications Opay, Jumia, Edozzier, ChatGPT (Dynamic Information)

Operating System

Software Solutions (Web Apps and Mobile Apps)

⇒ Three Tier Architecture



To understand how these tiers interact, consider a user signing up for a service:

1. **Request:** The user enters their details on the signup form (Presentation Tier).
2. **Processing:** The system validates the input and checks existing records (Logic Tier).
3. **Storage:** The user's information is securely saved (Data Tier).
4. **Response:** The system confirms the registration and redirects the user to the login or dashboard page.

1. Presentation Tier (Client Tier) – The Interface

This is the front-end layer that users directly interact with, such as a web browser or mobile app. Using a restaurant analogy, this functions as the waiter—they take your order, communicate it to the back-end, and present the final results to you.

2. Logic Tier (Application Tier) – The Processing Engine

This is the core of the application where business logic is executed. It acts as the kitchen—receiving requests from the presentation tier, processing inputs, performing necessary calculations, and coordinating data operations.

3. Data Tier (Database Tier) – The Data Storage

This is where all information is stored and managed. It functions as the restaurant's pantry and storage—holding essential data like customer records, inventory, and order history, which the Logic Tier accesses to fulfill requests.

⇒ Web Development Tools Installation and setups (Browsers, Text/Code Editors)

Web Development: Setting Up Your Environment

To begin your journey in web development, you need the right tools to build, test, and debug your projects effectively. A modern web browser and a reliable code editor are essential.

1. Web Browsers

A web browser is your window to the web and your primary tool for testing. Google Chrome is highly recommended due to its robust development suite.

Google Chrome

I. **Why use it:** It includes powerful Developer Tools (DevTools) for real-time debugging, performance analysis, and device simulation.

II. **Installation:**
    A. Visit the Google Chrome website.
    B. Download the installer for your operating system (Windows, macOS, or Linux).
    C. Run the installer and follow the on-screen prompts.
III. **Pro Tip:** Set Chrome as your default browser in your system settings to streamline your development workflow.

2. Text Editors (Code Editors)

A powerful code editor is crucial for writing clean, efficient code. **Visual Studio Code (VS Code)** is the industry standard.

Visual Studio Code

I. **Why use it:** It is free, fast, lightweight, and features excellent syntax highlighting, auto-completion, and native Git integration. Its extensive marketplace allows you to customize your workflow.
II. **Installation:**
    A. Visit the Visual Studio Code website.
    B. Download the version compatible with your operating system.
    C. Launch the installer and follow the instructions to complete the setup.

Introduction to HTML

⇒ What you need for the class

1. Your full attention
2. Your PC
3. Extension box (with plug)
4. Internet (Occasionally for now)

⇒ History and Development of HTML, Versions

I. Early Development: HTML was created by Tim Berners-Lee in 1989 at CERN (European Organization for Nuclear Research) to enable the sharing of documents over the World Wide Web (WWW).
II. 1991 - HTML 1.0: The first version, used for simple web pages with text and links.
III. 1995 - HTML 2.0: Introduced forms and tables.
IV. 1997 - HTML 3.2: Added support for design elements like fonts and colors.
V. 1999 - HTML 4.01: Separated content from design (CSS introduced).
VI. 2014 - HTML5: The latest version, supporting audio, video, and better structure for web apps.

There is no software solution that is currently at its peak

⇒ HTML (HyperText Markup Language) is the standard language used to create web pages. It is the backbone of most web content, providing structure and meaning to documents. HTML uses a system of tags to define the structure of web pages, allowing browsers to interpret and render them for users to view. HTML is the foundation of web development.



Steps in building a house

1. Already Bought a plot of land
2. Draw the plan - Architectural plan, survey, Urban ….
3. Foundation - Structure (My Decking) - HTML
4. Roof, fix windows, Paint, Furnish- CSS
5. Electrifying and Automation. - Javascript

⇒ Files and Folders

HTML
CSS
INTRO TO JS
BOOTSTRAP
VERSION CONTROL SYSTEM
HOSTING AND DEPLOYMENT

- HTML files are saved as .html

  An image (.png/jpeg/jpg), audio(.mp3) and a video(.mp4)

- Websites should be organized into folders for better structure; each project should own one folder to itself, to make your work well structured and to avoid confusion in the long run.

⇒ Steps to Creating an HTML project

- Create a folder and open in vs code (cmd)
- Initialize your html project (html:5/ doc/ !)

Assignment 1: Recreate an HTML project and initialize it.

⇒ HTML Syntax

An HTML element is defined by a start tag, some content, and an end tag:

Element = tag + content + attribute

<h1></h1> = tag

Hello, World! = content

<opening tag> Content goes here... </closing tag>

Example

```
<h1>Hello, World!</h1>
```

⇒ **Types of tags**

1. Container/Paired tags

These are tags that have an **opening tag** and a **closing tag**, enclosing content between them. Eg Hello Web! , <h1>Hello World!</h1>. E.t.c

2. Self-Closing/Empty tags

These tags do not have a closing tag; they **stand alone** and do not wrap around content.

Eg
, <hr>, <meta>, <input type="text">. E.t.c

Knowing that an Element contains an opening tag, your content and then the closing tag; there is something we call attribute.

An attribute is what goes inside your opening tag that gives characteristics to your element.

Assignment 2: List and test 80 more tags each and categorize them under closing and self closing tags.

⇒ **Element Levels**

1. Block-Level Elements

A block-level element in HTML is an element that starts on a new line and takes up the full width of its parent container.

Examples: , <p>, <h1> - <h6>, <section>, <article>, <ul>, <table>

2. Inline-Level Elements

An inline element is an element that does not start on a new line and only takes up as much width as necessary. It is used for styling or linking small parts of a document.

Examples: , <a>, <b>, <i>, , <img>

Assignment 3: List and test 20 more elements each and categorize them under block level and inline level elements.

⇒ **Semantic tags in HTML**

are elements that clearly describe their meaning both to the browser and to humans reading the code.

Meaningful tags clearly describe their purpose.

```
Tag Purpose
<header>    Represents the header section of a webpage.</header>
<nav>   Defines a navigation menu.</nav>
<section>   Groups related content together.</section>
<article>   Represents independent content like a blog post or news article.</article>
<aside> Contains side content (e.g., a sidebar).</aside>
<footer>    Defines the footer of a webpage.</footer>
<main>  The main content of a webpage.</main>
```

Using meaningful tags makes a web page easier to read and understand.

Example:

```
<header>Website Header</header>
<section>Main Content</section>
<footer>Page Footer</footer>
```

⇒ Tree Structure, Nesting and Indentation are very important in writing HTML

Commenting of codes also help in the readability of your code, it can help you save some information inside of your code but won't display on your web browser

⇒ To keep your hands on deck, you can do yourself good to watch resources on your portal. This will help you to be more enlightened and it will open your mind towards next classes.

Assignment 4: Use all you've learnt in HTML so far to build a simple web page about yourself, and your first day at SQI College of ICT.

⇒ **HTML LIST**

In HTML, lists are used to organize and display content in a structured way. There are three main types of lists:

There are three types of list in HTML

Ordered List : An ordered list is used when the order of items matters. Each item in the list is automatically numbered.

Here is how to write Ordered List

```html
<!-- Ordered List -->
<ol>
    <li>Wake up early</li>
    <li>Exercise</li>
    <li>Eat breakfast</li>
</ol>
```

Outcome

1. Wake up early
2. Exercise
3. Eat breakfast

Unordered List : An unordered list is used when the order of items does not matter. Items are displayed with bullet points.

Here is how to write Unordered List

```html
<!-- Unordered List -->
<ul>
    <li>Milk</li>
    <li>Eggs</li>
    <li>Bread</li>
</ul>
```

Outcome

- Milk
- Eggs
- Bread

Descriptive List : A description list is used for defining terms and their descriptions, like in a dictionary.

Here is how Descriptive List is written

```
<!-- Descriptive List -->
<dl>
    <dt>HTML</dt>
    <dd>HyperText Markup Language - the structure of web pages.</dd>

    <dt>CSS</dt>
    <dd>Cascading Style Sheets - used for styling web pages.</dd>
</dl>
```

Outcome

HTML

HyperText Markup Language - the structure of web pages.

CSS

Cascading Style Sheets - used for styling web pages.

Assignment 5: Use ordered list to write your best foods, then use unordered list to write the ingredients that makes up one of your best food and then use descriptive list to write the steps in preparing that your best food (where step 1, step 2... are your <dt>).

⇒ **HTML LINKS**

This is a very essential part of every web page we are building

There are four different ways

1. Linking to another website - links that takes you to another website
2. Linking to another page on the same website -links that takes you to another page on the same website
3. Linking to another part on the same web page- links that takes you to another part on the same web page
4. Functional links- (email, phone number) it takes you to an application on your device instead of taking you to another website or web page.

edu.sqi.ng

Note that HTML is the building block and then CSS is will modify it to make it look good

Assignment 6: Using the new things that you've learnt in HTML, rebuild a portfolio for yourself that can navigate to various pages like about you, your services and your contact information, also add functional links to your contact us page also. Include move to the top on all your pages.

⇒ **HTML FORMS**

In HTML, form is used to collect users information and send the information to the server. "Form" from the sentence "fill this form"

Form validation - This is how you check if the form's data is correct before submitting it. It helps catch errors before sending data to the server.

required

Form Submission

When a form is submitted, it sends the input data to a URL (specified in the action attribute)

Action - Submit "Where it should be submitted to"

Method - How it should be submitted

HTML Form controls

These are the various types of inputs and elements you can use in a form:
Form and Input tags - forms, input, button, select…

Label(<label for="the id of the input">) : This helps to instruct and guide what goes into the input.

Text Field (<input type="text">): Allows users to type text.

Password Field (<input type="password">): Hides the user's input (e.g., for passwords).

Email Field (<input type="email">): Validates if the input looks like an email.

Number Field (<input type="number">): Allows only numeric input.

Buttons (<button>): Triggers form submission or other actions.

Checkbox (<input type="checkbox">): Lets the user select one or more options.

Radio Buttons (<input type="radio">): Lets the user select one option from multiple choices.

Range(<input type="range">): A range bar for grading/level checking.

Dropdown (<select>): A drop-down list of options.

Text Area (<textarea>): A larger box for multi-line text.

⇒ **HTML MULTIMEDIA**
We have 3 multimedia tags in HTML
Images

Video

Audio

Controls, Autoplay and width

Assignment 7: Build a blog site containing Video, Audio, Images and Texts.

⇒ **HTML TABLES**

At the end of the class Students should have been able to create: Ludo game, Chess Board, Scrabble, Draft, Monopoly anything that tables can be used to create on the web page.

⇒ HTML tables allow web developers to arrange data into rows and columns.

⇒Tables are made up of two main components, which are mainly:

1. Columns - Vertical
2. Rows - Horizontal

Table tag won't do anything without contents which are

1. Rows in a table - tr(table-row)
2. Column in a table -td (table data)

A table won't show border naturally, here are several ways you can add border to your table:

1. By adding an attribute (border="1") to your table tag.
2. By using css to add border and we'll arrive the same place step one took us too
3. By adding border collapse to arrive at a very nice table and padding

⇒ Class example 1

S/N	Name	Account balance
1	David	$10,000,000
2	Seun	$20,000,000
2	Tolu	$30,000,000

⇒ Class work - Create a Meal time table for a week

Day	Breakfast	Lunch	Dinner
Monday	Rice	Bread	Amala
Tuesday	Beans	Garri	Wheat
Wednesday	Pap	Noddles	Eba
Thursday	Semovita	Rice	Spagghetti
Friday	Ofada-Rice	Potato	Pounded yam
Saturday	Fried- Rice	Chicken & Chips	Beans
Sunday	Bread	Rice and Beans	Pasta

⇒ Here is how to differentiate the title of of your columns, to make it more nice th-(table header) it will have a bolder font, making it the head of the table.

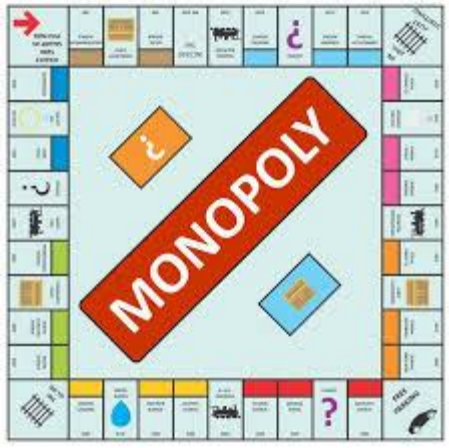
⇒ If i want to merge two rows you make use of row-span

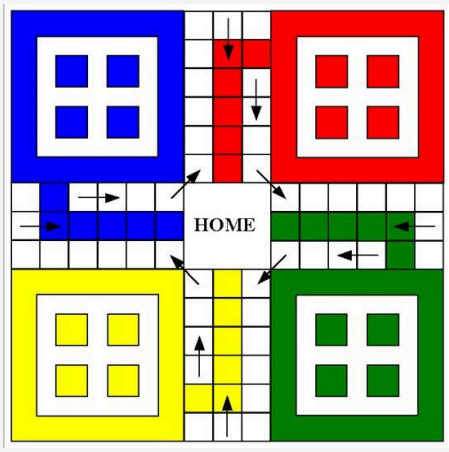
If i want to merge two columns together, you make use of col-span

The value imputed will determine the amount of column that the data should span through. (to make the data fall in the center of the two columns it is spanning through, you can use text align =center in css).

Assignment 8

Build any of the following with HTML tables







Assignment 9

Build facebook login and signup page and connect them together.

Introduction to Version Control System

MODULE 1: INTRODUCTION TO GIT

History of Git (Global Information Tracker) and GitHub

Types of Version Control Systems (Distributed and Centralized)

What is it used for :

- Collaboration
- Version Control History

- Portfolio
- Open Source
- Backup
- Remote

Git Installation and set up

MODULE 2: GIT SETUP, INIT, STAGE AND SNAPSHOT

git init - to initialize an empty repository

git config (username and email):

- git config –-global user.name “Your Name”
- git config --global user.email “Your email”

Stages in git

git add . - To add all files to the staging area

git rm – cached name of file

git commit -m “commit message”

git log

git status,

git diff,

git reset

MODULE 3: GIT BRANCH, MERGE AND IGNORE

Using touch to create files

.gitignore(how to ignore files)

git branch: how to create a branch, what it means.

git checkout

git merge

How to work in a branch etc (checkout into a branch) etc.

Providing a copy of git cheat sheet

Note: There is more to git!!. Thanks!!!

HOSTING AND DEPLOYMENT

What You Will Learn

MODULE 1: HOSTING AND DEPLOYMENT

Definition of common terms (Hosting, Deployment, Domain Name, Servers).

Deploying to any hosting platform. netflify.com is advised.

Changing domain names and deleting deployments.

How to redeploy an already deployed app

Providing a list of free and paid hosting platforms.

List of sites

Paid:

1. Namecheap
2. Qservers
3. Godaddy
4. Whogohost
5. Bluehost
6. hostinger

Free:

1. 000webhost

2. infinityfree
3. netlify
4. Firebase
5. GitHub pages
6. Heroku
7. Vercel

Introduction to CSS

MODULE 1: INTRODUCTION TO CSS

➔ The power of CSS in web designing : This helps to modify our web pages to our taste, we can control/instruct our element to display how we want using CSS.



➔ How does CSS look like (CSS Syntax): CSS has a very simple syntax which is

property: value;

style=”property: value;” used for inline

selector {

property: value;

}

Used for both internal and external

➔ Methods of linking HTML and CSS together

Inline = style attribute (style=””)

Internal = style tag (<style></style>)
External = style file (style.css)

➔ Definitions: Selectors, Properties, Values, and Units

Selectors: these are what we use to call upon the element we want to style or beautify on our page.

Types of Selectors

1. Simple - element, class, and id
2. Combinator
3. Pseudo

Property: these are the commands of the style you want to give your element (color)

Value: these are answers to our property (the red/blue/brown that answers to color command, in the property)

Unit: these tells the size of the value we are giving e.g. 10px for text size, 10% for width e.t.c.

Assignment 10

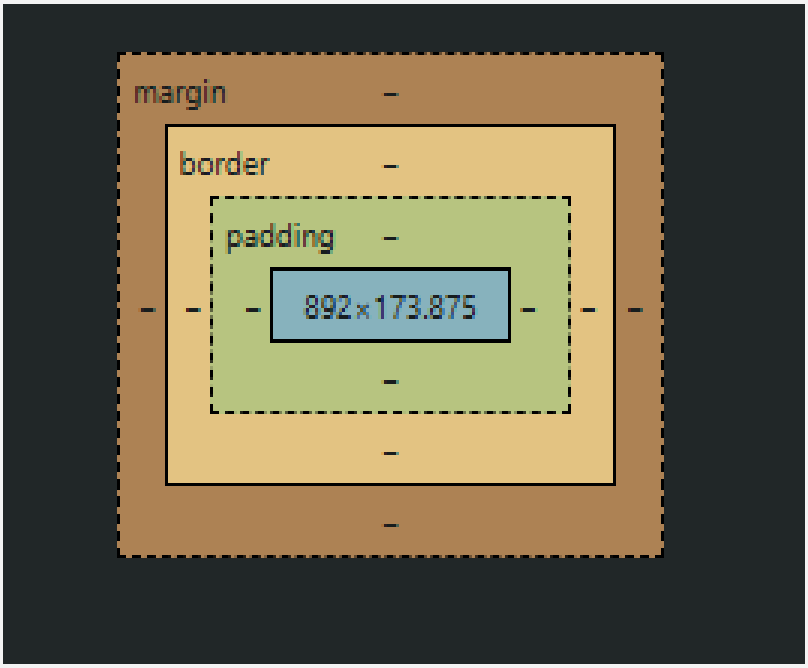
Update all your HTML projects by adding CSS to them.

MODULE 2: CSS BOX MODEL

- ➔ Passing Borders, Margin, and Padding to HTML Tags
- ➔ Box Model
- ➔ Box model comprises of 4 major elements as seen in the image in our browser
- ➔ These elements are
- ➔ 1. Margin - Carton Brown - This simply is the spacing without, i.e the spacing between that element and another element
- ➔ 2. Padding - Orange - This is the opposite of Margin, Meaning it is the spacing within, also known as the space between content and Border
- ➔ 3. Border - Green - This simply means the boundary of out content (wrapping both the content and the padding)
- ➔ 4. Content - Skyblue - This is what is written from your HTML as the content of your element.
- ➔
- ➔
- ➔ top right bottom and left
- ➔
- ➔
- ➔ padding: 20px; for all sides
- ➔ padding: 20px 30px; first value for vertical and the second value for horizontal
- ➔ padding: 20px 30px 40px ; padding: top right bottom right;

➔ padding: 20px 30px 40px 50px;

➔



`

Border: a boundary that gives a limit to an element on our page.

Padding: is the spacing between the border and the content of the element(spacing within).

Margin: is the spacing outside the border of an element (spacing without).

➔ Specifying height and width for HTML elements

Percentage values

width: 100%;

height: 100%

Viewport values

width: 100vw;

height: 100vh;

Pixels values

width: 100px;

height: 100px;

➔ Scope/Level of HTML elements (Block and Inline)

display: block; (makes the element a block level element)

display: inline; (makes the element an inline level element)

Assignment 11

Build an Hero Page with linear-gradient image on the background and have h1, p, and a button that links you to your portfolio (the one you built with html).

MODULE 3: CSS TEXT & FONT FORMATTING

➔ Font-families (generic, Font Names)
➔ Font CDN (Google Fonts, Adobe Fonts)
➔ Font size, Font style, Font Weight, etc
➔ Text decoration, line height, letter spacing, word spacing, etc.
➔ Text transform,text align, text-indent,text-shadow (text-shadow: 2px 2px 5px black ;)
➔ Color: hex code, names, RGB, RGBA.
➔

Assignment 12

Clone this site Giant Robots: thoughtbot Blog leaving out the form, navbar and the footer in it.

MODULE 4: WEB PAGE LAYOUT

The Display property (inline, block, inline-block, flex, grid)

Display: flex;

justify-content, align-items, flex-direction, flex-wrap, gap and other flex properties are used in laying out the web page and making it neat and well aligned.

➔ Background Images, Background Colors, Gradients, Box Shadow

Structuring Web page using Grid

Assignment 13

Clone this site https://web.dev leaving nothing out.

Position is also used to layout your web page, it simply means aligning an element to a part or place on the web page.

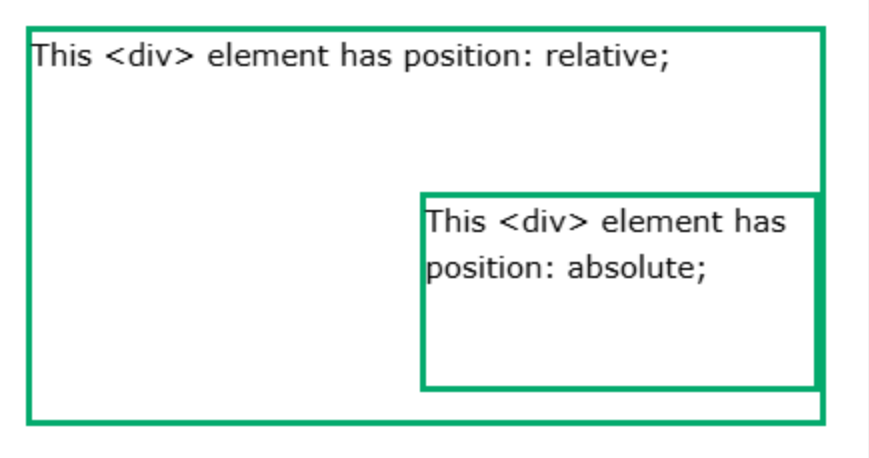
Static- Default position for any element.

Sticky- this makes the element scroll and then become fixed.

Fixed- Positioning something on your page to not be moveable i.e fixed.

Relative- Parent, being able to relate with the child

Absolute- child, being able to relate with the parent



Setting to coordinate - top, right bottom left.

Z-index- allows a element to display over another element

Assignment 14

Clone this site https://wikipedia.org leaving nothing out on the page.

MODULE 5: CSS 3 CORE

Transforms (rotate, translate and scale)

Transition, Filters

Animation values

Name -This is the **name of the keyframes** you want to use.(box)

Duration - How long the animation should take to complete **one cycle**.(2s)

Timing Function - This defines the **speed curve** of the animation.(linear, ease, ease-in ease-out, ease-in-out)

Delay - How long to **wait before starting** the animation.(1s)

Iteration Count -How many times the animation should run(1, 2, infinite).

Direction- The direction of the animation cycle.(normal, reverse, alternate(forward then backwards), alternate-reverse)

Fill-mode - Controls how the element is styled **before and after the animation** runs:

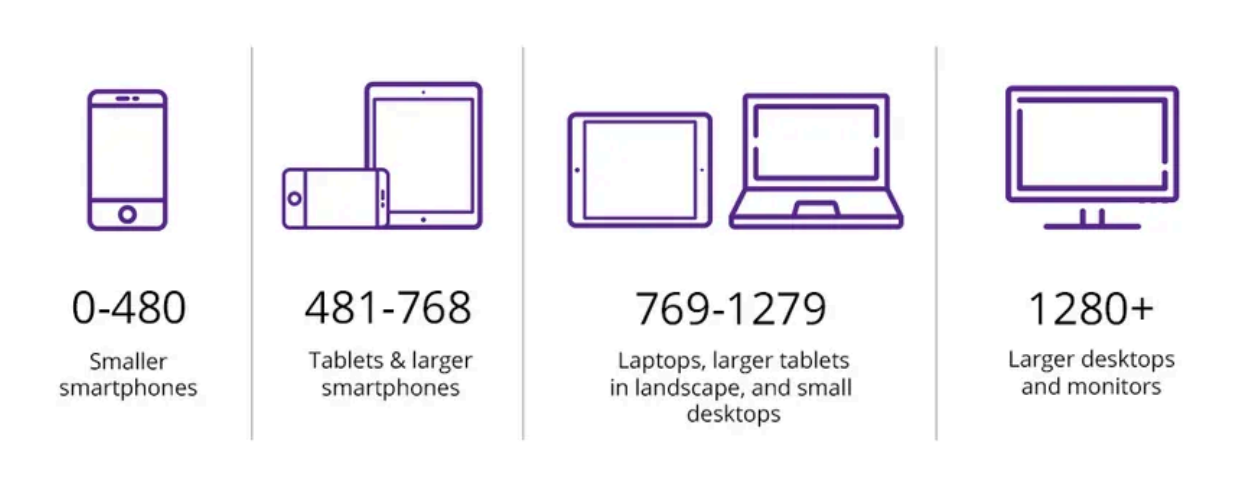
None default; element does not retain styling

Forwards; keeps styles after animation ends

Backwards: applies styles before animation starts

Both - combines both

MODULE 6: RESPONSIVE WEB DESIGN



How to design a website that behaves well on all screens (phone, tablet and

desktop) using media queries,

Feature Queries

Styling for print

BOOTSTRAP CSS

MODULE 1: INTRODUCTION TO BOOTSTRAP CSS

- ➔ CSS libraries (Bootstrap, tailwind, materialize, e.t.c)- refers to collections of css classes, It help us to make our page responsive without stress
- ➔ History of Bootstrap CSS, Versions https://getbootstrap.com
- ➔ How is Bootstrap different from conventional CSS
- ➔ Bootstrap Installation:
    1. Download
    2. CDN
- ➔ Bootstrap Basic Examples

MODULE 2: BOOTSTRAP UTILITIES AND TYPOGRAPHY

Reboot, Typography, Background, Borders,
Colors, Images, Table,Figures,
Display, Flex, Float, Interactions
Overflow, Position, Shadows, Sizing,
Spacing, Text, Vertical align, Visibility

MODULE 3: BOOTSTRAP COMPONENTS

Accordion, Alerts, Badge, Breadcrumb
Buttons, Button group, Card, Carousel
Close button. Collapse, Dropdowns
List group, Modal, Navs & tabs
Navbar, Pagination, Popovers

Progress, Scrollspy, Spinners

Toasts, Tooltips

MODULE 4: BOOTSTRAP LAYOUT

Breakpoints, Bootstrap Containers classes

Making Web Layout using Bootstrap Grid System

Columns, Gutters, Utilities, Z-index

MODULE 5: BOOTSTRAP FORMS

Overview, Form control, Select,

Checks & radios, Range, Input group

Floating labels, Form Layout, Form Validation

MODULE 6: PUTTING IT TOGETHER

Build a responsive modern website using Bootstrap

13 Sites To Get Free Illustrations for Web developers

- blush. design
- drawkit. io
- humaaans. com
- icons8. com
- iconscout. com
- illustrations. co
- iradesign. io
- isometric .online
- manypixels. co

- openpeeps. com
- pixeltrue. com
- storyset. com
- undraw. co

Other useful links

1. remove.bg (to freely remove background from pictures)

2. undraw.co (for free illustrations and sketches. You'll like it)

3. blush.design (similar to undraw but very great as you can modify the images or even animate them. Not totally free but let us cultivate the habit of buy cool things that can help is in our career)

4. unsplash.com (picture repository just like Pinterest)

5. dribble.com (designers and creatives hub where you can find cool and crazy prototypes to replicate)

6. carrd.co (simple, free, and fully responsive one page sites for pretty much anything. You can use it for your portfolio)

7. flourish.studio (for animated data presentation. Might need a little JS though)

For personal development

8. untools.co (helps you organize your thinking patterns)

9. Lumen5.com (AI video creator)